A Lock Anyone Can Close, but Only You Can Open

Explain it in your own words

Use this guide to understand each animation, choose your phrasing, and keep a natural YouTube delivery. Every clip has a matching frame, the idea behind the movement, and a specific delivery cue.

16 clips | 09:44.7 continuous video | 10 guide pages

A simple speaking rhythm

1. Set up: Tell viewers what the objects represent and ask one small question.

2. Follow the change: Point to what moves, grows, disappears or gets replaced.

3. Explain the cause: Say why that change happened. Use the mechanism notes as your checklist.

4. Let it land: Pause on the result, then connect it to the next question.

Use the timeline flexibly

The next page gives exact clip boundaries in the current continuous MP4. Individual clip files begin at 00:00. In the self-paced player, a phase holds until you advance, so your own recording can be longer or shorter. These timestamps describe the video; delivery cues describe how you might explain it.

Press Right / Left to move between phases, Up / Down to adjust speed, R to replay, and Shift+P for the separate script window. Rehearse the cause-and-effect in your own words before recording.

Open this video in the local player | Full narration reference | Sources

The complete timeline

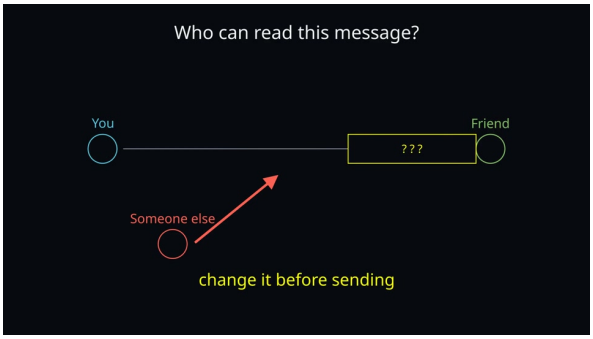
Find the clip, see the question being explained, and jump to its guide page. Times are MM:SS.t in the full MP4.

CLIP / TIME	WHAT YOU ARE EXPLAINING	PAGE
01 00:00.0 - 00:36.5	Who can read this message?	3
02 00:36.5 - 01:13.1	A lock with two different jobs	3
03 01:13.1 - 01:47.4	Keep what is left over	4
04 01:47.4 - 02:24.8	Build the number fifty-five	4
05 02:24.8 - 03:02.7	Find a useful repeating structure	5
06 03:02.7 - 03:38.4	The public rule repeats a multiplication	5
07 03:38.4 - 04:14.5	Choose a matching private rule	6
08 04:14.5 - 04:50.1	Turn seven into thirteen	6
09 04:50.1 - 05:25.8	Keep the numbers manageable	7
10 05:25.8 - 06:02.8	Bring the message back	7
11 06:02.8 - 06:40.2	Why the pair fits together	8
12 06:40.2 - 07:15.9	Try the wrong private number	8
13 07:15.9 - 07:52.4	What must remain hard?	9
14 07:52.4 - 08:30.3	The tiny example needs boundaries	9
15 08:30.3 - 09:06.8	Connect the two kinds of rules	10
16 09:06.8 - 09:44.7	Rebuild the idea from the start	10

Use the clip number as your anchor when recording. Matching frames show the completed diagram; watch the movement too, because the change is often the explanation.

CLIP 01 / Who can read this message?

00:00.0 - 00:36.5 in full video | 36.5s clip



What the viewer sees
A private message becomes unreadable while an observer looks on.

What you need to explain - and why
Encryption transforms readable information so an unintended observer cannot understand it. The puzzle is how to send privately before sharing a secret.

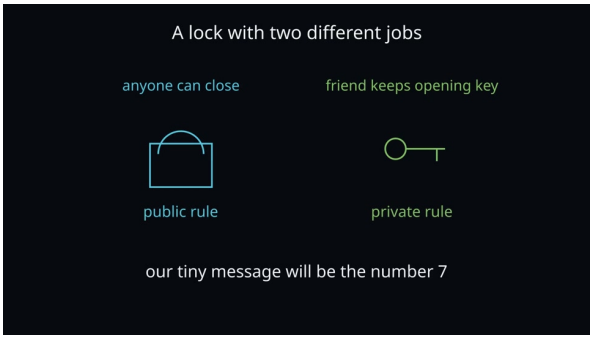
Tone and emphasis
Begin with why privacy is needed, not with equations.

Next: A lock with two different jobs

Open clip 01 | My phrasing: _____

CLIP 02 / A lock with two different jobs

00:36.5 - 01:13.1 in full video | 36.5s clip



What the viewer sees
An open padlock and a separately kept opening key appear.

What you need to explain - and why
The public operation can protect a message while the matching private operation recovers it. Messages are represented as numbers for the calculation.

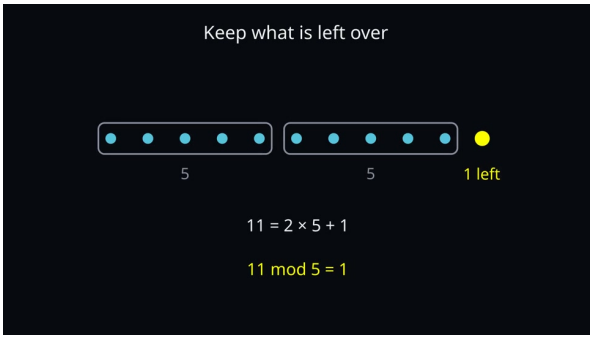
Tone and emphasis
Use the lock analogy, then make clear that arithmetic does the actual work.

Next: Keep what is left over

Open clip 02 | My phrasing: _____

CLIP 03 / Keep what is left over

01:13.1 - 01:47.4 in full video | 34.3s clip



What the viewer sees
Eleven counters become two groups of five plus one leftover.

What you need to explain - and why
Modulo means keep the remainder after removing complete groups. Eleven mod five is one.

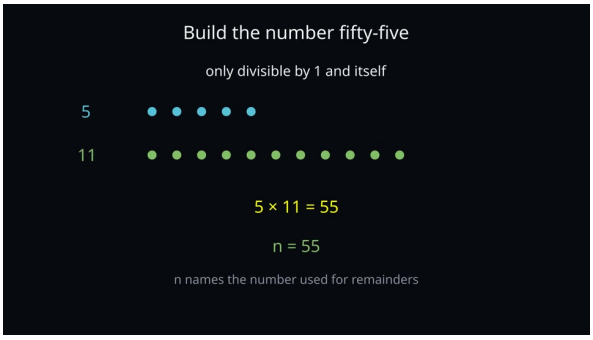
Tone and emphasis
Count real dots first and pronounce mod only after explaining leftovers.

Next: Build the number fifty-five

Open clip 03 | My phrasing: _____

CLIP 04 / Build the number fifty-five

01:47.4 - 02:24.8 in full video | 37.4s clip



What the viewer sees
Five and eleven are shown as prime building blocks.

What you need to explain - and why
Primes divide evenly only by one and themselves. Their product 55 becomes n, the number used for remainder calculations.

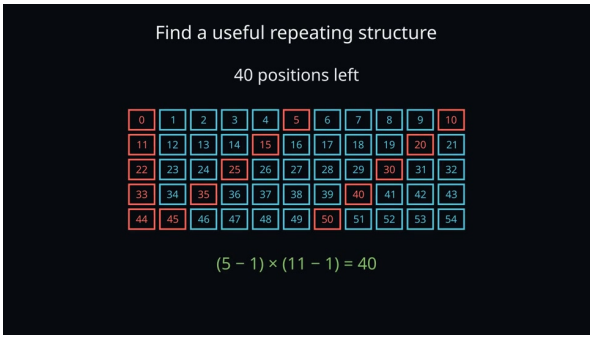
Tone and emphasis
Say that the tiny numbers are chosen to make the mechanism visible.

Next: Find a useful repeating structure

Open clip 04 | My phrasing: _____

CLIP 05 / Find a useful repeating structure

02:24.8 - 03:02.7 in full video | 37.9s clip



What the viewer sees
Multiples of five and eleven are crossed out in a 0-54 grid.

What you need to explain - and why
Forty positions share no factor above one with 55. This count, (5 - 1)(11 - 1), supports the matching-key construction.

Tone and emphasis
Explain what was counted; do not say every message has a cycle of exactly forty.

Next: The public rule repeats a multiplication

Open clip 05 | My phrasing: _____

CLIP 06 / The public rule repeats a multiplication

03:02.7 - 03:38.4 in full video | 35.7s clip



What the viewer sees
The public pair 55 and 3 is shown with repeated multiplication.

What you need to explain - and why
The public exponent e = 3 means multiply three copies of the message, then keep the remainder modulo 55. Three is compatible with the count forty.

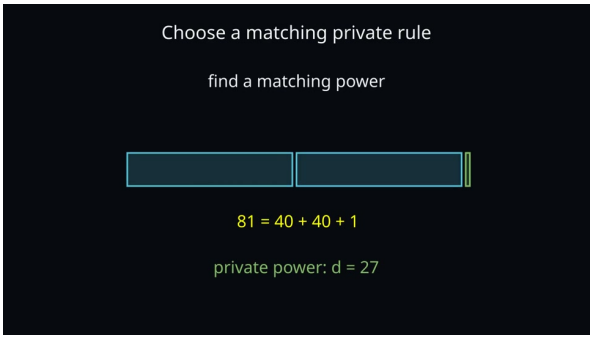
Tone and emphasis
Define cubing as three copies multiplied together.

Next: Choose a matching private rule

Open clip 06 | My phrasing: _____

CLIP 07 / Choose a matching private rule

03:38.4 - 04:14.5 in full video | 36.1s clip



What the viewer sees
A private exponent is chosen using groups of forty.

What you need to explain - and why
d = 27 fits because 3 times 27 = 81 = 2 times 40 + 1. The private operation is related to the public construction.

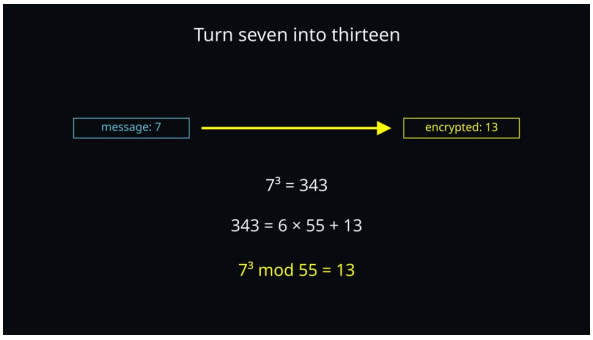
Tone and emphasis
Read the arithmetic as the test that makes these two powers match.

Next: Turn seven into thirteen

Open clip 07 | My phrasing: _____

CLIP 08 / Turn seven into thirteen

04:14.5 - 04:50.1 in full video | 35.7s clip



What the viewer sees
Seven is cubed and reduced to thirteen.

What you need to explain - and why
7 times 7 times 7 = 343. Removing six groups of 55 leaves 13, the ciphertext in our tiny example.

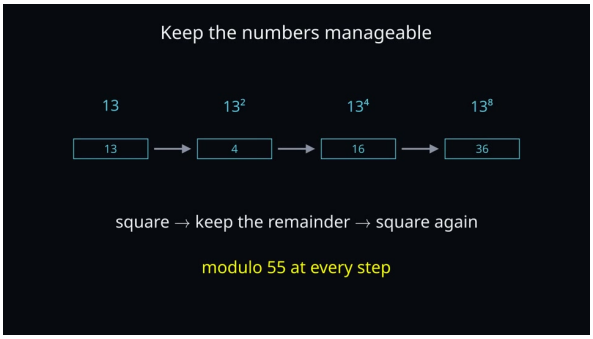
Tone and emphasis
Let viewers calculate the leftover before revealing the encrypted number.

Next: Keep the numbers manageable

Open clip 08 | My phrasing: _____

CLIP 09 / Keep the numbers manageable

04:50.1 - 05:25.8 in full video | 35.7s clip



What the viewer sees
Repeated squaring keeps remainders small.

What you need to explain - and why
13 squared leaves 4 modulo 55; squaring again gives 16, then 36. Intermediate remainders retain the needed result without writing the enormous full power.

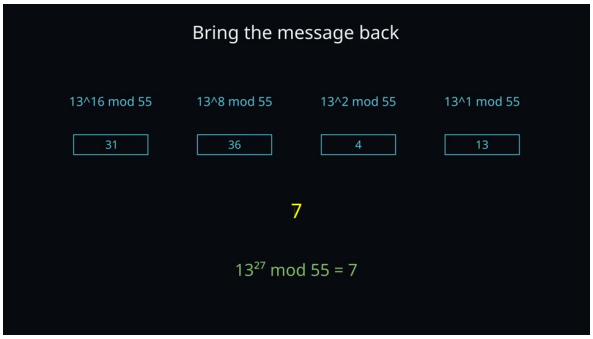
Tone and emphasis
Present one small calculation at a time rather than reading a wall of powers.

Next: Bring the message back

Open clip 09 | My phrasing: _____

CLIP 10 / Bring the message back

05:25.8 - 06:02.8 in full video | 37.0s clip



What the viewer sees
Selected power remainders combine to recover seven.

What you need to explain - and why
27 = 16 + 8 + 2 + 1. Multiplying the corresponding remainders gives 13 to the power 27 modulo 55, which returns 7.

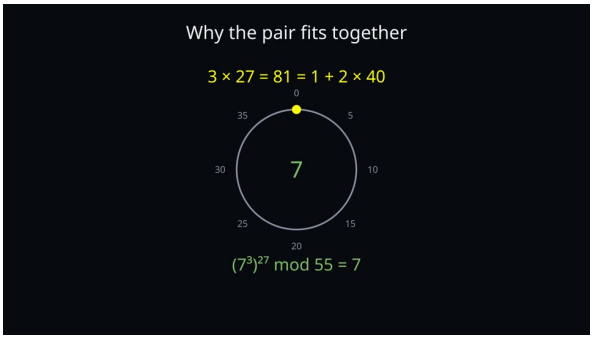
Tone and emphasis
Trace the selected boxes, then pause on the recovered original message.

Next: Why the pair fits together

Open clip 10 | My phrasing: _____

CLIP 11 / Why the pair fits together

06:02.8 - 06:40.2 in full video | 37.4s clip



What the viewer sees
A repeating arithmetic picture links the two exponents.

What you need to explain - and why
Encryption then decryption combines exponents into 81. The matching modular construction makes recovery work; the circle illustrates repetition rather than a universal forty-step cycle.

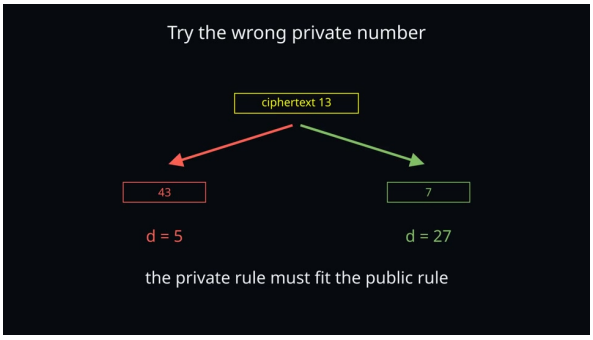
Tone and emphasis
Give the reason for the pairing without attempting a long number-theory proof.

Next: Try the wrong private number

Open clip 11 | My phrasing: _____

CLIP 12 / Try the wrong private number

06:40.2 - 07:15.9 in full video | 35.7s clip



What the viewer sees
A guessed private power gives forty-three instead of seven.

What you need to explain - and why
13 to the power 5 modulo 55 equals 43. An arbitrary private number usually fails because the operations must be correctly related.

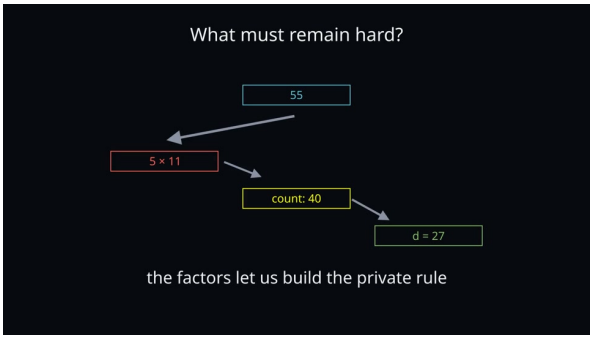
Tone and emphasis
Use the wrong guess to test correctness, not as a proof of security.

Next: What must remain hard?

Open clip 12 | My phrasing: _____

CLIP 13 / What must remain hard?

07:15.9 - 07:52.4 in full video | 36.5s clip



What the viewer sees
The product and its hidden prime factors are contrasted.

What you need to explain - and why
Knowing 5 and 11 lets someone reconstruct the private-key calculation. Real RSA relies on much larger carefully chosen numbers and hard mathematical problems.

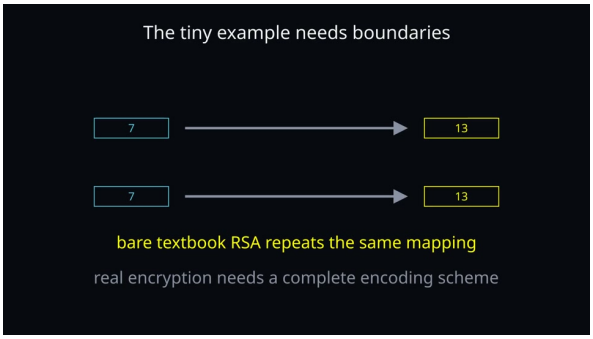
Tone and emphasis
State clearly that factoring fifty-five is easy and protects nothing.

Next: The tiny example needs boundaries

Open clip 13 | My phrasing: _____

CLIP 14 / The tiny example needs boundaries

07:52.4 - 08:30.3 in full video | 37.9s clip



What the viewer sees
Repeated bare encryption produces a repeated result.

What you need to explain - and why
The raw toy rule is deterministic, which can leak structure. Practical encryption needs a complete encoding and padding scheme with appropriate randomness.

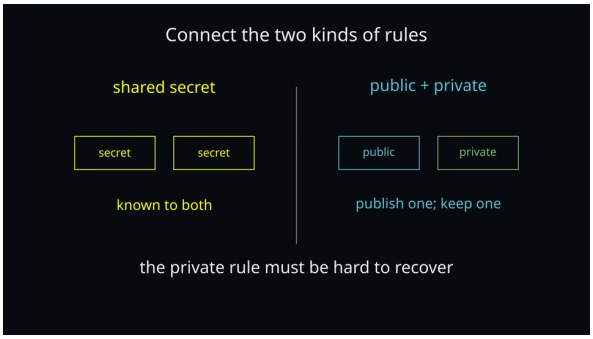
Tone and emphasis
Separate understanding the mathematical mechanism from building secure software.

Next: Connect the two kinds of rules

Open clip 14 | My phrasing: _____

CLIP 15 / Connect the two kinds of rules

08:30.3 - 09:06.8 in full video | 36.5s clip



What the viewer sees
Shared-secret and public/private roles are compared.

What you need to explain - and why
A shared-secret arrangement needs a suitable secret in common. A public-key arrangement publishes one operation while keeping its recovery operation private.

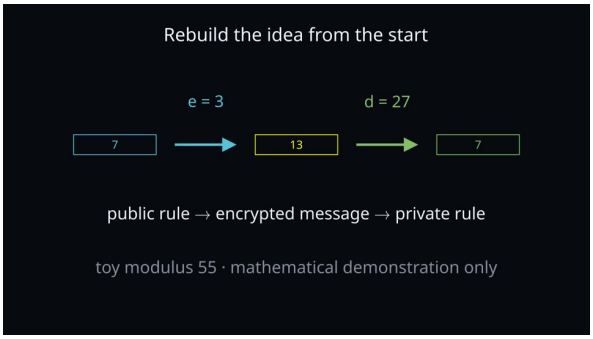
Tone and emphasis
Return to the original communication problem after the arithmetic.

Next: Rebuild the idea from the start

Open clip 15 | My phrasing: _____

CLIP 16 / Rebuild the idea from the start

09:06.8 - 09:44.7 in full video | 37.9s clip



What the viewer sees
Seven becomes thirteen publicly and returns privately.

What you need to explain - and why
The public pair (55, 3) and matching private exponent 27 demonstrate paired operations. Remainders keep results in a limited range.

Tone and emphasis
Close with the journey of one message and the original public-lock idea.

Close: return to the opening question and its answer.

Open clip 16 | My phrasing: _____